

CPPDescent

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Chapter 1

CPP-Descent

Part of the “Software Development for Computing Systems” course of the Department of Informatics and Telecommunications, UoA for the first semester of the academic year 2023-2024.

A library that implements the KNN-Graph creation and search algorithms described in [1], [2].

1.1 Dependencies

For local development, you will need Make and CMake. To install them on an apt-based linux distro, run the following commands:

```
$ sudo apt update && sudo apt upgrade
$ sudo apt install make
$ sudo apt install cmake
```

1.1.1 Minor dependencies

- The project uses Google Test for its testing needs. Gtest is used as a git submodule of this repo, that is downloaded by CMake the first time you clone. That means that nothing extra needs to be installed to run the tests.
- If you would like to generate the documentation, you will need doxygen installed.

1.2 Code Structure

In the [wiki](#), you can find documentation generated by Doxygen. It is also available in [pdf form](#).

1.2.1 Code Coverage

This project uses LCOV for its code coverage needs. The results are uploaded upon commit [here](#).

1.3 Contributors

- [Konstantinos Chousos](#) (1115202000215)
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1.4 Bibliography

[1] W. Dong, C. Moses, and K. Li, “Efficient k-nearest neighbor graph construction for generic similarity measures,” in *Proceedings of the 20th international conference on World wide web*, in WWW '11. New York, NY, USA: Association for Computing Machinery, Mar. 2011, pp. 577–586. doi: [10.1145/1963405.1963487](#).

[2] “How PyNNDescent works — pynndescent 0.5.0 documentation.” Accessed: Oct. 10, 2023. [Online]. Available: https://pynndescent.readthedocs.io/en/latest/how_pynndescent_works.html

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 3

File Index

3.1 File List

Here is a list of all documented files with brief descriptions:

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Chapter 4

Class Documentation

4.1 cppdescent Class Reference

A sample function for the cppdescent class, used to test the correctness of the CMake files.

```
#include <cppdescent.hpp>
```

Public Member Functions

- int [hello](#) (void)
Prints "Hello World".

4.1.1 Detailed Description

A sample function for the cppdescent class, used to test the correctness of the CMake files.

4.1.2 Member Function Documentation

4.1.2.1 hello()

```
int cppdescent::hello (  
    void )
```

Prints "Hello World".

Returns

int exit code

The documentation for this class was generated from the following files:

- [cppdescent.hpp](#)
- [cppdescent.cpp](#)

4.2 List Class Reference

ADT [List](#).

```
#include <ADTList.hpp>
```

Public Member Functions

- [List](#) (DestroyFunc value=nullptr)
Construct a new [List](#) object.
- [~List](#) ()
Destroy the [List](#) object.
- void [insertNext](#) ([ListNode](#) *node, Pointer value)
Adds a node after the given node.
- void [removeNext](#) ([ListNode](#) *node)
Removes the node after the given node.
- Pointer [find](#) (Pointer value, CompareFunc compare)
Finds the first value equal to the value parameter.
- DestroyFunc [setDestroyValue](#) (DestroyFunc value)
Set the Destroy Value object to a new value.
- int [getSize](#) ()
Get the size of the [List](#).
- int [increaseSize](#) ()
Increase the list size.
- int [decreaseSize](#) ()
Decrease the list size.
- [ListNode](#) * [getHead](#) ()
Getter for the head of the [List](#).
- [ListNode](#) * [getTail](#) ()
Getter for the tail of the [List](#).
- [ListNode](#) * [next](#) ([ListNode](#) *node)
Returns the node after the given one.
- Pointer [nodeValue](#) ([ListNode](#) *node)
Returns the value of the node.
- [ListNode](#) * [findNode](#) (Pointer value, CompareFunc compare)
Finds the first node that has value equal to the value parameter.

4.2.1 Detailed Description

ADT [List](#).

An Abstract Data Type [List](#) with linear access to data, and addition and removal to/from arbitrary positions.

4.2.2 Constructor & Destructor Documentation

4.2.2.1 List()

```
List::List (
    DestroyFunc value = nullptr )
```

Construct a new [List](#) object.

Parameters

<i>destroyValue</i>	If <i>destroyValue</i> != nullptr, the <i>destroyValue</i> function will be called each time a node is removed.
---------------------	---

head is a virtual node that is always present in the list.

Parameters

<i>destroyValue</i>	The destroy function for the values of the list. Defaults to nullptr.
---------------------	---

4.2.2.2 ~List()

```
List::~~List ( )
```

Destroy the [List](#) object.

If the *destroyValue* function is not nullptr, it is called for every value.

4.2.3 Member Function Documentation**4.2.3.1 decreaseSize()**

```
int List::decreaseSize ( )
```

Decrease the list size.

Returns

int The updated size.

4.2.3.2 find()

```
Pointer List::find (
    Pointer value,
    CompareFunc compare )
```

Finds the first value equal to the value parameter.

Find the first value equal to the value parameter.

Parameters

<i>value</i>	The value based on which the search is done.
<i>compare</i>	A pointer to the function that will compare the node values.

Returns

Pointer A pointer to the resulting node.

Parameters

<i>value</i>	The value to search for.
<i>compare</i>	A pointer to the compare function.

Returns

Pointer A pointer to found value.

4.2.3.3 findNode()

```
ListNode * List::findNode (
    Pointer value,
    CompareFunc compare )
```

Finds the first node that has value equal to the value parameter.

Find the first node with value equal to the value parameter.

Parameters

<i>value</i>	The value to search.
<i>compare</i>	The compare function to use.

Returns

ListNode* A pointer to the resulting node.

Parameters

<i>value</i>	The value to search for.
<i>compare</i>	A pointer to the compare function.

Returns

ListNode* A pointer to the found node.

4.2.3.4 getHead()

```
ListNode * List::getHead ( )
```

Getter for the head of the [List](#).

Get the first element of the [List](#).

Returns

ListNode* The head.

Returns

ListNode* Pointer to the first element.

4.2.3.5 getSize()

```
int List::getSize ( )
```

Get the size of the [List](#).

Get the size of the list.

The size is updated upon entry/removal of nodes, so the retrieval of the size is O(1).

Returns

int

Returns

int The size.

4.2.3.6 getTail()

```
ListNode * List::getTail ( )
```

Getter for the tail of the [List](#).

Get the last element of the list.

Returns

ListNode* The tail.

If the list is empty, the virtual head is the tail, so we return nullptr.

Returns

ListNode* Pointer to the last element.

4.2.3.7 increaseSize()

```
int List::increaseSize ( )
```

Increase the list size.

Returns

int The updated size.

4.2.3.8 insertNext()

```
void List::insertNext (
    ListNode * node,
    Pointer value )
```

Adds a node *after* the given node.

Insert a new element after the node.

Parameters

<i>node</i>	The node that the new node will be placed after.
<i>value</i>	The value of the new node.

Parameters

<i>node</i>	The node after which the element is inserted.
<i>value</i>	The value of the new element.

4.2.3.9 next()

```
ListNode * List::next (
    ListNode * node )
```

Returns the node after the given one.

Get the next node.

Parameters

<i>node</i>	
-------------	--

Returns

ListNode* A pointer to the next node.

Parameters

<i>node</i>	
-------------	--

Returns

ListNode*

4.2.3.10 nodeValue()

```
Pointer List::nodeValue (  
    ListNode * node )
```

Returns the value of the node.

Get the node's value.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer A generic pointer to the value.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer

4.2.3.11 removeNext()

```
void List::removeNext (  
    ListNode * node )
```

Removes the node *after* the given node.

Removes the node after the given node.

Parameters

<i>node</i>	The node before the node to be removed.
-------------	---

Parameters

<i>node</i>	The node after which the element is removed.
-------------	--

4.2.3.12 setDestroyValue()

```
DestroyFunc List::setDestroyValue (
    DestroyFunc value )
```

Set the Destroy Value object to a new value.

Set the destroy value function.

Parameters

<i>value</i>	A pointer to the new destroy function.
--------------	--

Returns

DestroyFunc A pointer to the old destroy function.

Parameters

<i>value</i>	The new destroy function.
--------------	---------------------------

Returns

DestroyFunc The old destroy function.

The documentation for this class was generated from the following files:

- ADTList.hpp
- [ADTList.cpp](#)

4.3 ListNode Class Reference

Node of a list.

```
#include <ADTList.hpp>
```

Public Member Functions

- [ListNode](#) ()
Construct a new [List](#) Node object that is empty.
- [ListNode](#) (Pointer value)
Construct a new [List](#) Node object with value 'value'.
- void [setNext](#) ([ListNode](#) *next)
Destroy the [List](#) Node object.
- [ListNode](#) * [getNext](#) ()
Getter for the next node.
- Pointer [getValue](#) ()
Getter for the value of the node.

4.3.1 Detailed Description

Node of a list.

A list node is represented by its value. It also holds a pointer to the next node in the list.

4.3.2 Constructor & Destructor Documentation

4.3.2.1 [ListNode\(\)](#) [1/2]

```
ListNode::ListNode ( ) [inline]
```

Construct a new [List](#) Node object that is empty.

4.3.2.2 [ListNode\(\)](#) [2/2]

```
ListNode::ListNode (
    Pointer value ) [inline]
```

Construct a new [List](#) Node object with value 'value'.

Parameters

<i>value</i>	a generic pointer to the value of the node.
--------------	---

4.3.3 Member Function Documentation

4.3.3.1 getNext()

```
ListNode * ListNode::getNext ( )
```

Getter for the next node.

Returns

ListNode* The pointer to the next node.

4.3.3.2 getValue()

```
Pointer ListNode::getValue ( )
```

Getter for the value of the node.

Returns

Pointer A generic pointer to the value.

4.3.3.3 setNext()

```
void ListNode::setNext (
    ListNode * next )
```

Destroy the [List](#) Node object.

Setter for the next node.

Parameters

<i>next</i>	The pointer to the next node.
-------------	-------------------------------

The documentation for this class was generated from the following files:

- ADTList.hpp
- [listNode.cpp](#)

Chapter 5

File Documentation

5.1 ADTList.hpp

```
00001
00011 #pragma once
00012 #include "common.hpp"
00013
00014 #define LIST_BOF (ListNode*)0
00015 #define LIST_EOF (ListNode*)0
00016
00024 class ListNode {
00025 public:
00030     ListNode() : next(nullptr), value(nullptr){};
00036     ListNode(Pointer value) : next(nullptr), value(value){};
00041     // ~ListNode();
00047     void setNext(ListNode* next);
00053     ListNode* getNext();
00059     Pointer getValue();
00060
00061 private:
00062     ListNode* next;
00063     Pointer value;
00064 };
00065
00073 class List {
00074 public:
00081     List(DestroyFunc value = nullptr);
00086     ~List();
00093     void insertNext(ListNode* node, Pointer value);
00099     void removeNext(ListNode* node);
00107     Pointer find(Pointer value, CompareFunc compare);
00114     DestroyFunc setDestroyValue(DestroyFunc value);
00123     int getSize();
00124     int increaseSize();
00125     int decreaseSize();
00131     ListNode* getHead();
00137     ListNode* getTail();
00144     ListNode* next(ListNode* node);
00151     Pointer nodeValue(ListNode* node);
00159     ListNode* findNode(Pointer value, CompareFunc compare);
00160
00161 private:
00162     DestroyFunc destroyValue;
00163     int size;
00164     ListNode* head;
00165     ListNode* tail;
00166 };
```

5.2 common.hpp

```
00001 #pragma once
00002
00003 typedef void* Pointer;
00004
00005 typedef int (*CompareFunc)(Pointer a, Pointer b);
00006
00007 typedef void (*DestroyFunc)(Pointer value);
```

5.3 cppdescent.hpp

```
00001
00006 class cppdescent {
00007 public:
00013     int hello(void);
00014 };
```

5.4 ADTList.cpp File Reference

Implementation of an ADT linked list.

```
#include "cppdescent/ADTList.hpp"
Include dependency graph for ADTList.cpp:
```

5.5 cppdescent.cpp File Reference

Implementation of cppdescent library.

```
#include "cppdescent/cppdescent.hpp"
#include <iostream>
Include dependency graph for cppdescent.cpp:
```

5.5.1 Detailed Description

Implementation of cppdescent library.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-27

Copyright

Copyright (c) 2023

5.6 listNode.cpp File Reference

Implementation of the listNode to be used with linkedList.

```
#include "cppdescent/ADTList.hpp"
Include dependency graph for listNode.cpp:
```

5.6.1 Detailed Description

Implementation of the listNode to be used with linkedList.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-29

Copyright

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